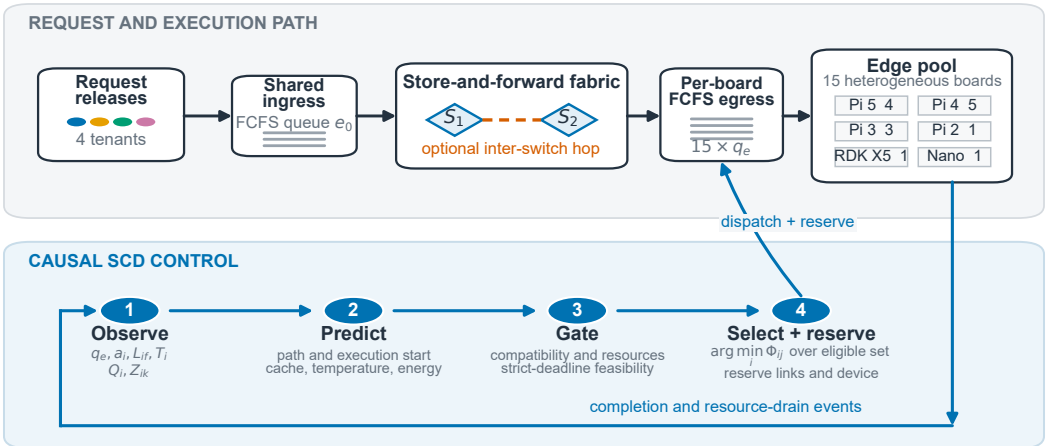

REQUEST AND EXECUTION PATH
Request releases
4 tenants
Shared ingress
FCFS queue e_0
Store-and-forward fabric
S_1
S_2
optional inter-switch hop
Per-board FCFS egress
15 × q_e
Edge pool
15 heterogeneous boards
Pi 5 4
Pi 4 5
Pi 3 3
Pi 2 1
RDK X5 1
Nano 1
dispatch + reserve
CAUSAL SCD CONTROL
1
Observe
q_e, a_i, L_{if}, T_i
Q_i, Z_{ik}
2
Predict
path and execution start
cache, temperature, energy
3
Gate
compatibility and resources
strict-deadline feasibility
4
Select + reserve
$\arg\min_i \Phi_{ij}$ over eligible set
reserve links and device
completion and resource-drain events